

BINGJIAN HUANG

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Dynamic Graphics Project Lab (DGP), Department of Computer Science, University of Toronto

EDUCATION

Ph.D. in Human-Computer Interaction

2020-Present

Dynamic Graphics Project Lab (DGP), University of Toronto, Ontario, Canada

Supervisor: Prof. Daniel Wigdor

B.E. in Computer Science and Technology — GPA 3.67/4.00

2016-2020

Tsinghua University, Beijing, China

Minor in Finance and Entrepreneurship — GPA 3.81/4.00

2017-2020

Tsinghua University, Beijing, China

PUBLICATIONS

AeroHaptix: A Wearable Vibrotactile Feedback System for Enhancing Collision Avoidance in UAV Teleoperation

Bingjian Huang, Zhecheng Wang, Qilong Cheng, Siyi Ren, Hanfeng Cai, Antonio Alvarez Valdivia, Karthik Mahadevan, Daniel Wigdor

Under Review

VibraForge: A Scalable Prototyping Toolkit For Creating Spatialized Vibrotactile Feedback Systems

Bingjian Huang, Siyi Ren, Yuewen Luo, Qilong Cheng, Hanfeng Cai, Yeqi Sang, Mauricio Sousa, Paul H Dietz, Daniel Wigdor

CHI 2025

Investigating the Effect of Intensity and Frequency on Vibrotactile Localization Accuracy

Bingjian Huang, Paul H. Dietz, Daniel Wigdor

IEEE Transactions on Haptics 2024

Cohabitant: The Design, Implementation, and Evaluation of a Virtual Reality Application for Interfaith Learning and Empathy Building

Mohammad Rashidujjaman, Reem Ayad, Ashratuz Zavin Asha, Bingjian Huang, Selin Okman, Dina Sabie, Hasan Shahid Ferdous, Robert Soden, Syed Ishtiaque Ahmed

CHI 2024

User Burden of Microinteractions: An In-lab Experiment Examining User Performance and Perceived Burden Related to In-situ Self-reporting

Xinghui Yan, Yuxuan Li, Bingjian Huang, Sun Young Park, Mark W Newman

MobileHCI 2021

Toward Lightweight In-situ Self-reporting: An Exploratory Study of Alternative Smartwatch Interface Designs in Context

Xinghui Yan, Shriti Raj, Bingjian Huang, Sun Young Park, Mark W Newman

IMWUT (UBICOMP) 2020

Designing and Evaluating Hand-to-Hand Gestures with Dual Commodity Wrist-Worn Devices

Yiqin Lu, Bingjian Huang, Chun Yu, Guanhong Liu, Yuanchun Shi

IMWUT (UBICOMP) 2020

RESEARCH EXPERIENCE

Graduate Research Assistant

2020.8-Present

Dynamic Graphics Project Lab (DGP), University of Toronto, Advisor: Prof. Daniel Wigdor

- Conducting psychophysics studies to investigate fundamental haptic perception, e.g., vibrotactile spatial acuity, phantom sensations, multisensory integration.

- Building open-source vibrotactile suit that supports over 120 vibrotactile actuators with low-latency and fine-grained control.
- Applying haptic feedback to various domains, including virtual reality, UAV teleoperation, and accessibility.

Undergraduate Research Assistant

2018.10-2019.11

Pervasive Computing Group, Tsinghua University, Advisors: Prof. Yuanchun Shi, Prof. Chun Yu

- Researched on hand-to-hand gesture design space exploration and smartwatch recognition.
- Interviewed 10 people and surveyed 24 people about gesture preference.
- Collected raw IMU sensor data, extracted features, trained SVM & LSTM Pytorch models.
- Developed a real-time gesture recognition system, validated in daily scenarios.
- Revised paper and added real-context experiments based on major revision review

Visiting Scholar and Research Intern

2019.7-2019.9

Interaction Ecologies Group, School of Information, University of Michigan, Ann Arbor, Advisors: Prof. Mark W. Newman, Prof. Sun Young Park

- Researched on in-situ low-burden self-reporting tools.
- Developed 11 Android smartwatch self-reporting prototypes to investigate different design concepts.
- Conducted user studies about scenarios and techniques with User Enactment Method.

Undergraduate Research Assistant

2018.3-2018.9

Pervasive Computing Group, Tsinghua University, Adviser: Prof. Yuanchun Shi, Prof. Chun Yu

- Developed a 3D Telepresence software framework aiming to reach high co-presence.
- Used CUDA to implement parallel algorithms such as TSDF and Marching Cubes.
- Conducted user experiments and interviewed participants

SKILLS

Programming Languages: C/C++/Python/Java

Programming Tools: MATLAB/Qt/UnityVR/CUDA/PyTorch/Android Studio/Raspberry Pi

Electronic Hardware: PCB Design/CAD Design/Embedded Design

Languages: English(TOFEL 115 (speaking: 25), GRE 159+170+3.5)

RESEARCH INTERESTS

VR/AR/MR, novel input methods and future user interfaces

Human Perception and Illusion and how they affects user experience in virtual environments

Wearable Computing with smartwatch, smart rings and smart clothes

HONORS AND AWARDS

Wolfond Fellowship (\$20,000, \$10,000 each year), University of Toronto *September, 2020*

GPHL Scholarship for Academic Excellence(Top 20%), Tsinghua University *September, 2019*

SOHU Scholarship for Academic Excellence(Top 10%), Tsinghua University *September, 2017*

2nd Prize, National Olympiad in Informatics in Provinces(NOIP) *November, 2014*

MISCELLANEOUS

Sports Enthusiast: Skiing, Basketball, High Diving, Hiking